Honors Geometry – 4.1 & 4.2 Practice **WORKED KEY**

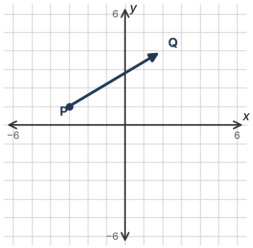
Translations and Reflections • Sections 4.1 – 4.2

Show your work. Label every image with primes (A′, A″). Each part ends with a self-check, so use it before you move on.

Part A — Translations (4.1)

A translation slides EVERY point the same way. Vector ⟨a, b⟩ is the rule (x, y) → (x + a, y + b). Ex: ⟨−2, 5⟩ moves (3, 1) to (1, 6).

1. Name the vector and write its component form.



→ **PQ→ (vector PQ); ⟨5, 3⟩**

from P(−3, 1) to Q(2, 4): right 5, up 3

2. Write a rule for the translation that moves A(2, −1) to A′(−3, 4).

(x, y) → (x + ____, y + ____)

→ **(x, y) → (x − 5, y + 5)**

x: 2 → −3 is −5; y: −1 → 4 is +5

3. Use (x, y) → (x + 4, y − 3) to find the images of J(−2, 5), K(0, 1), and L(3, −2).

→ **J′(2, 2), K′(4, −2), L′(7, −5)**

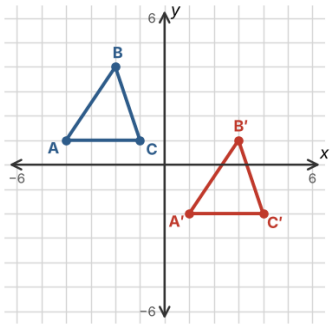
4. P′(6, −1) is the image of P after a translation by ⟨−2, 5⟩. Find P.

Undo it: go the opposite way.

→ **P(8, −6)**

undo ⟨−2, 5⟩ with ⟨2, −5⟩: (6 + 2, −1 − 5)

5. Graph △ABC with A(−4, 1), B(−2, 4), C(−1, 1) and its image after a translation by ⟨5, −3⟩.



→ **A′(1, −2), B′(3, 1), C′(4, −2)**

6. Translate by (x, y) → (x + 3, y − 1), then by (x, y) → (x − 5, y + 4). Write ONE rule that does both.

→ **(x, y) → (x − 2, y + 3)**

add the moves: 3 + (−5) = −2; −1 + 4 = 3

✓ **Check yourself:** Add all six coordinates of your images in #3. You should get 8.

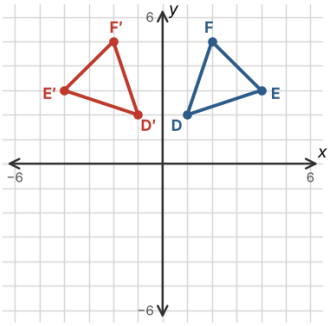
Part B — Reflections (4.2)

x-axis: (x, y) → (x, −y) y-axis: (x, y) → (−x, y) y = x: (x, y) → (y, x) y = −x: (x, y) → (−y, −x). For x = h or y = k: count to the line, then go the SAME distance past it.

7. Reflect the point (3, −5) in each line.

→ **x-axis (3, 5); y-axis (−3, −5); y = x (−5, 3); y = −x (5, −3)**

8. Graph △DEF with D(1, 2), E(4, 3), F(2, 5) and its reflection in the y-axis.



→ **D′(−1, 2), E′(−4, 3), F′(−2, 5)**

9. Reflect M(−2, 3) in the line x = 1.

M is ____ units left of x = 1, so M′ is ____ units right.

→ **M′(4, 3)**

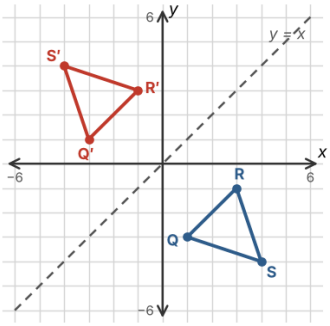
3 units left of x = 1 → 3 units right

10. Reflect N(4, −1) in the line y = 2.

→ **N′(4, 5)**

3 units below y = 2 → 3 units above

11. Graph △QRS with Q(1, −3), R(3, −1), S(4, −4) and its reflection in the line y = x. Draw the line first.



→ **Q′(−3, 1), R′(−1, 3), S′(−4, 4)**

swap x and y

12. A(−3, 2) maps to A′(2, −3). Name the line of reflection.

→ **y = x**

the coordinates swapped: (x, y) → (y, x)

✓ **Check yourself:** Average each point and its image in #9 and #10. You should land exactly on the line of reflection: x = 1 and y = 2.

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Part C — Glide reflections and compositions

Glide reflection: a translation, THEN a reflection in a line parallel to the translation. One step at a time: write the middle image (′) before the final image (″).

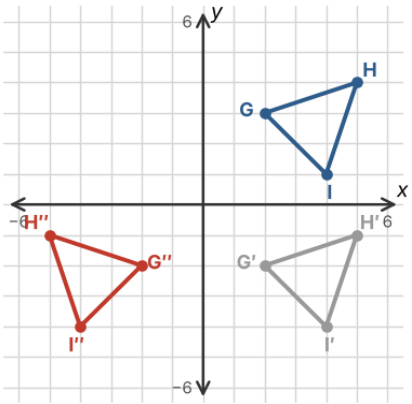
13. Glide reflection: translate (x, y) → (x + 4, y), then reflect in the x-axis. Find the image of T(−3, 2).

T′ (____, ____) → T″ (____, ____)

→ T″(1, −2)

T′(1, 2) → T″(1, −2)

14. Graph △GHI with G(2, 3), H(5, 4), I(4, 1). Translate by ⟨0, −5⟩, then reflect in the y-axis. Label both images.



→ G″(−2, −2), H″(−5, −1), I″(−4, −4)

middle: G′(2, −2), H′(5, −1), I′(4, −4)

15. Reflect P(2, 5) in the x-axis, then translate by ⟨3, 0⟩. Now do it in the OTHER order. Do you get the same point?

→ (5, −5) both ways: yes

reflect first: (2, −5) → (5, −5). translate first: (5, 5) → (5, −5). The line is parallel to the slide, so order doesn't matter.

16. Reflect (1, 3) in the line x = 2, then reflect that image in the line x = 6. Where does it end up? What single transformation does the same thing?

→ (9, 3); a translation by ⟨8, 0⟩

x = 2 → (3, 3); x = 6 → (9, 3). Two parallel mirrors = a slide of TWICE the gap (2 · 4 = 8).

✓ **Check yourself:** In #15 both orders must match. If they don't, redo the reflection step, not the slide.

Part D — Push further

Same moves, hiding behind a sentence.

17. Error analysis. Marco reflected (4, −2) in the y-axis and got (4, 2). Name his mistake in one sentence. Then give the correct image.

→ (−4, −2)

He reflected in the x-axis (changed y). In the y-axis, x changes sign: (−4, −2).

18. A translation maps △ABC to △A′B′C′. AB = 7 cm and m∠B = 40°. Find A′B′ and m∠B′. Explain why in one sentence.

→ A′B′ = 7 cm, m∠B′ = 40°

Translations are rigid motions: they keep every length and every angle.

19. How many lines of symmetry does each figure have?

→ (a) 6 (b) 2 (c) 1 (d) 0

a parallelogram that is not a rectangle or rhombus has none

20. Reflect P(2, 4) in the line y = −3. Then write a rule for reflecting ANY point (x, y) in the line y = −3.

Hint: the line is halfway, so the y-values must average to −3.

→ P′(2, −10); (x, y) → (x, −6 − y)

P is 7 above y = −3 → 7 below: −10. (y + y′)/2 = −3 → y′ = −6 − y

✓ **Check yourself:** Your four answers in #19 should add up to 9. In #20, test your rule on P: it has to give P′.